AP Precalculus Review Guide

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Range: Chapter 1-5, 7-11, 13

1.6 Solving quadratic equations

- ***Quadratic Equation***
- ✓ **Standard Form:** A quadratic equation in one variable:

  $ax^2+bx+c=0$, $a\neq 0$ and a,b,c are real numbers.
- ✓ **Solution** or **Root:** Substituting each solution for the variable can make the equation true.
- ✓ **Method:** ① *factoring* ② *completing the square* ③ *the quadratic formula*

- ***The Discriminant*** b^2-4ac

In the quadratic formula, $x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$, the radicand b^2-4ac is called the discriminant of the equation.

- If $b^2-4ac>0$, the equation has two real solutions.
- If $b^2-4ac=0$, the equation has one real solution.
- If $b^2-4ac<0$, the equation has zero real solutions. The solutions are imaginary numbers.

- ***Choosing a Method of Solution***

Situation	Method to use
$a,b,$ and c are integers and b^2-4ac is a perfect square.	factoring
The equation has the form $x^2+(\text{even number})x+\text{constant}=0$	completing the square

1.7 Quadratic functions and their graph

✔ **Quadratic Function**

The graph of $f(x)=ax^2+bx+c$ (another form $y=a(x-h)^2+k$), where $a\neq 0$, is a **Parabola**.

Axis: The graph has a vertical axis of symmetry: $x=-\frac{b}{2a}$

Vertex: Point where the parabola intersects the axis of symmetry: $(-\frac{b}{2a}, f(-\frac{b}{2a}))$

Opening:

If $a>0$, opens **upward**.

If $a<0$, opens **downward**.

Maximum or Minimum:

If $a>0$, vertex is the **minimum**.

If $a<0$, vertex is the **maximum**.

End behaviors:

If $a>0$, function **increases** without bound.

If $a<0$, function **decreases** without bound.

Zeros of the function:

A quadratic can have 0, 1, or 2 real solutions depending on b^2-4ac.

Graph can have 0, 1, or 2 **x-intercepts** (real zeros).

Axis of symmetry intersects x-axis at the **midpoint** of the segment between x-intercepts.

Ø Rate of change and Monotonicity

A **positive** rate of change indicates that as one quantity increases or decreases, the other quantity does the same, and the graph is **increasing or decreasing throughout the interval** (i.e., **monotonic**).

A **negative** rate of change indicates that as one quantity increases, the other quantity decreases, as one quantity decreases, the other quantity increases, and the graph is **not monotonic** (i.e., it **changes direction**).

Ø Rate of change and Concavity

When the average rate of change over equal-length input-value intervals is increasing for small-length intervals, the graph of the function is **concave up**.

When the average rate of change over equal-length input-value intervals is decreasing for small-length intervals, the graph of the function is **concave down**.

Hint:

You can also determine the concavity of quadratic functions by graphs. That is,
when the quadratic function is opening **upward**, it is **concave up**.
When the quadratic function is opening **downward**, it is **concave down**.

2.6 Solving polynomial equations by factoring

- **Rational Root Theorem**

Use factoring to solve the general polynomial equation,

$$a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + a_0 = 0$$

$$(qx - p)(\qquad\qquad\qquad\qquad) = 0$$

If $qx - p$ is a factor (so that $x = \frac{p}{q}$ is a root), then q must divide a_n, and p must divide a_0.

> Let $P(x)$ be a polynomial of degree n with integral coefficients and a nonzero constant terms:
>
> $$P(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + a_0, \text{ where } a_0 \neq 0$$
>
> If one of the roots of the equation $P(x) = 0$ is $x = \frac{p}{q}$ where p and q are nonzero integers with no common factor other than 1, then p must be a factor of the **constant term** a_0 and q must be a factor of the **leading coefficient** a_n.

3.2 Polynomial inequalities in one variable

➢ ***Polynomial Inequalities***

Let $P(x)$ be any polynomial. Then $P(x) < 0$ and $P(x) > 0$ are called **polynomial inequalities**.

Method 1 Use a sign graph of $P(x)$.

Method 2 Analyze a graph of $P(x)$.

$P(x) < 0$ when the graph is *below* the x-axis;

$P(x) > 0$ when the graph is *above* the x-axis.

3.4 Linear programming

➢ ***The Corner-Point Principle***

A maximum or minimum value of a linear expression $P = Ax + By$, if it exists, will occur at a corner point of the feasible region.

-- The principle applies only to **convex** polygonal regions.

-- **Convex:** every line segment joining two points of the region is contained in the region.

- Linear Programming: A mathematical model consisting of a linear function to be optimized (maximized or minimized) and a set of inequalities representing the restrictions on resources;

- Objective Function: The function to be maximized or minimized in a linear program; This function might represent amounts of money earned or spent, goods manufactured or delivered, or any of a variety of other quantities.

- Constraints: The linear inequalities;

- Feasible Region: The solution of the system of linear inequalities; If a linear program has a solution, it will occur at a vertex (corner point) of the feasible region. If there are multiple solutions, at least one of them will occur at a vertex.

- To solve a linear program:

- Solve the system of linear inequalities. Determine the coordinates of the vertices of the feasible region. That is, solve the system of equations consisting of the lines whose intersection is the vertex.

- Evaluate the objective function to be optimized at each vertex.

- Identify the maximum or minimum value of the objective function.

Ø Note that if two vertices of the feasible region are both solutions to a linear program, every point on the line segment connecting the two vertices will also be a solution.

4.2 Operations on functions

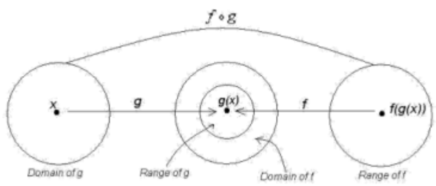
- ***The Composite of Functions***

Another method of combining two functions is by successive application of the functions in a specific order. This binary operation is called **composition of functions**.

Given two functions f and g, the **composite function** $f \circ g$ is defined by $(f \circ g)(x) = f(g(x))$ and is read "f circle g of x equals f of g of x". The domain of $(f \circ g)(x)$ is the set of elements x in the domain of g such that $g(x)$ is in the domain of f.

- ✓ In the composition $f \circ g$, the domain of f must contain the range of g.
- ✓ Sometimes, restrict the domain of g to make $g(x)$ in the domain of f.
- ✓ The composition "$f \circ g$" is the operation of applying g and then f.

  The composition "$g \circ f$" is the operation of applying f and then g.

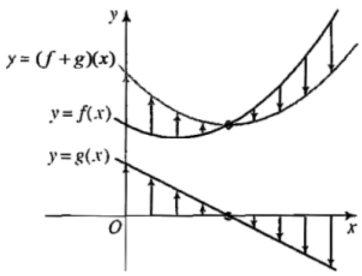


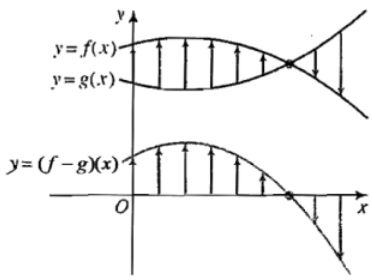
- ***The Sum, Difference, Product, and Quotient of Functions***

Given two functions f and g, new functions $f+g, f-g, f \cdot g,$ and f/g can be formed by adding, subtracting, multiplying, and dividing.

The DOMAIN of each resulting function consists of those values of x <u>common</u> to the domains of f and g. The domain of the <u>quotient</u> function is further restricted by excluding all values that make $g(x)=0$.

Sum	$(f+g)(x) = f(x) + g(x)$
Difference	$(f-g)(x) = f(x) - g(x)$
Product	$(f \cdot g)(x) = f(x) \cdot g(x)$
Quotient	$(f/g)(x) = \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)},\ g(x) \neq 0$





4.5 Inverse functions

➢ ***Inverse Functions***

Definition: A function f has an **inverse function** g if and only if

(i) $f(g(x)) = x$ for all x in the domain of $g(x)$ and

(ii) $g(f(x)) = x$ for all x in the domain of $f(x)$.

(If two function are inverses of each other, their composition give the identity function $h(x) = x$.)

Notation: The inverse function of a function f is denoted f^{-1} or $f^{-1}(x)$. $\boxed{(f \circ f^{-1})(x) = (f^{-1} \circ f)(x) = x}$.

(Note that f^{-1} is NOT the reciprocal of the function f. The reciprocal of f, $\frac{1}{f(x)}$, is denoted $[f(x)]^{-1}$.)

-- A function $y = f(x)$ that has an inverse is called a **one-to-one function**, because not only does each x-value correspond to exactly one y-value, but also each y-value corresponds to exactly one x-value.

-- For every point (a,b) on the graph of $y = f(x)$, the point (b,a) is on the graph of $y = f^{-1}(x)$. The graphs of a function and its inverse are ***symmetric*** with respect to the line $y = x$.

4.8 Partial fractions, rational functions

1.	To factor the denominator	$x^2 - 2x - 3 = (x-3)(x+1)$
2.	Express as two fractions using A and B for the numerators.	$\frac{7x-13}{x^2-2x-3} = \frac{A}{x-3} + \frac{B}{x+1}$
3.	Multiply by the least common denominator.	$7x - 13 = A(x+1) + B(x-3) = (A+B)x + (A-3B)$
4.	Solve for A and B.	① $\begin{cases} 7 = A + B \\ -13 = A - 3B \end{cases}$; ② $\begin{cases} x = 3: 7\cdot 3 - 13 = A(3+1) + B\cdot 0 \Rightarrow A = 2 \\ x = -1: 7\cdot(-1) - 13 = A\cdot 0 + B\cdot(-1-3) \Rightarrow B = 5 \end{cases}$
5.	Express with partial fractions	$\frac{7x-13}{x^2-2x-3} = \frac{2}{x-3} + \frac{5}{x+1}$

- In partial fraction decomposition, a linear factor $(x-a)$ is repeated n times. If $(x-a)^n$ is a factor of the denominator, then the partial fraction decomposition must include

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_n}{(x-a)^n}$$

➢ **Limit of a function**

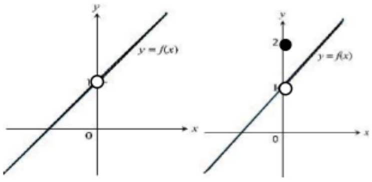
When x is getting closer and closer to c, $f(x)$ is approaching to a fixed number L.Then we can say "the limit of $f(x)$, as x approaches c, is L".

$$\lim_{x \to c} f(x) = L$$

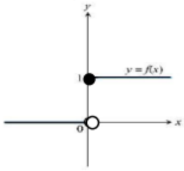
If so, then *f*(*x*) can be made to be as close to *L* as desired by making *x* sufficiently close to *c*.

- **Three basic kinds of discontinuity at a point**

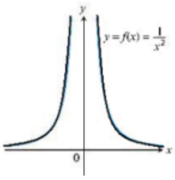
I. Point

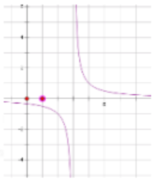


II. Jump



III. Infinite





- **Horizontal asymptote(s) i.e. end behavior asymptote(s)**

  If $\lim_{x\to\infty} f(x) = b$, or $\lim_{x\to-\infty} f(x) = b$, then line $y = b$ is a horizontal asymptote of $f(x)$.

- **Vertical asymptote(s)**

  If $\lim_{x\to a^-} f(x) = \pm\infty$, or $\lim_{x\to a^+} f(x) = \pm\infty$, then line $x = a$ is a vertical asymptote of $f(x)$.

➤ Rational Functions

If $P(x)$ and $Q(x) \neq 0$ are polynomial functions, then

$$f(x) = \frac{P(x)}{Q(x)}$$

is a **rational function** (quotient of two polynomials).

-- The graphs of rational functions frequently display **vertical** and **removable** discontinuities.

Chapter 4 Functions

Asymptote Rules (To Find End Behavior Asymptotes)

Let $\frac{f(x)}{g(x)}$ be a rational function where $f(x)$ is a polynomial of degree n and $g(x)$ is a polynomial of degree m.

If $g(a) = 0$ and $f(a) \neq 0$, then $x = a$ is a **vertical** asymptote.

Three possible conditions determine a **horizontal** asymptote:

- • If $n < m$, then $y = 0$ is a horizontal asymptote.
- • If $n > m$, then there is **no** horizontal asymptote. There are **slant or curve** asymptotes. The end behavior asymptote is the quotient polynomial function that is found when the polynomial $P(x)$ is divided into $Q(x)$. If that polynomial is linear, the rational function has a slant asymptote parallel to the graph of the line.
- • If $n = m$, then $y = c$ is a horizontal asymptote, where c is the quotient of the leading coefficients of f and g.

4.9 The binominal theorem

- **The Binomial Theorem**

  If n is a positive integer, then:

  $(a+b)^n\ a^n b^0 + na^{n-1}b^1 + \frac{n(n-1)}{2\cdot 1}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3\cdot 2\cdot 1}a^{n-3}b^3 + \frac{n(n-1)(n-2)(n-3)}{4\cdot 3\cdot 2\cdot 1}a^{n-4}b^4 \ldots + a^0 b^n$ T

  he denominators of these coefficients involve the product of the integers from n down to 1, we can write these in a shorter form using **factorial notation**.

- Factorial of integers

  When r is a positive integer, $r!$ means the product of all the positive integers from r down to 1.

The binomial coefficient of each term $\frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$ can be denoted by $\binom{n}{r}$.

So, $\binom{n}{r} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$

It can be simplified to $\frac{n!}{r!(n-r)!}$.

To summarize, the binomial theorem states that, if n is a positive integer,

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots \binom{n}{r}a^{n-r}b^r + \cdots \binom{n}{n}b^n$$

5.3 Exponential functions

- **Exponential Function with base b**

- The basic form of the graph of an exponential function $f(x)=b^x$ when $b>1$.
  - -- The point $(0,1)$ is common to all such graphs.
  - -- The value of b determines their "steepness".
  - -- As x approaches $-\infty$, the distance between the curve and x-axis → 0. Therefore, the exponential function is **asymptotic** to the x-axis. ***Horizontal asymptote***: x-axis
  - -- Since b^x approaches ∞ as x approaches ∞, the range of the exponential function is $\{y : y>0\}$. And the function is increasing and one-to-one.
- In general, any exponential function $f(x)=c^x$ is **one-to-one**.
- The graph of an exponential function can be reflected, dilated or translated.

- **Application**

→ Consider a community with an initial population P.

Suppose the rate of increase in population of the community for each year is represented by r, where $0<r<1$.

-- After 1 year, the population of the community: $P+rP$ *or* $P(1+r)$;

-- After 2 years, the population of the community: $P(1+r)+rP(1+r)$ *or* $P(1+r)^2$;

-- After 3 years, the population of the community: $P(1+r)^3$;

...After t years, the population of the community: $P(1+r)^t$.

→ Exponential growth and decay are often written in the forms such as

1) $A(t)=A_0(1+r)^t$ $\quad$ $A_0=$ amount at time $t=0$ and $r=$ growth rate
2) $A(t)=A_0 b^{t/k}$ $\quad$ $k=$ time needed to multiply A_0 by b

→ **The Rule of 72** If a quantity is growing at r% per year (or month), then the ***doubling time*** is approximately $(72\div r)$ years (or months).

5.6 Law of logarithms

- **Laws of Logarithms**

If M, N and b are positive real numbers, b is a positive number other than 1, then

1. $\log_b MN=\log_b M+\log_b N$;

2. $\log_b \dfrac{M}{N}=\log_b M-\log_b N$

3. $\log_b M=\log_b N$ if and only if $M=N$

4. $\log_b M^k=k\log_b M$, for any real number k

5.7 Exponential equations, changing bases

Since exponential functions are one-to-one, $b^m = b^p$ if and only if $m = p$.

- **Exponential Equations:** equations that contain one or more exponential expressions.
  → express the exponential expressions in terms of the same base;
  → use algebraic techniques such as factoring.

- Logarithm are especially useful for solving exponential equations in which both sides of the equation cannot be written in terms of the same base, for $\log_b M = \log_b N$ if and only if $M = N$.

-- To evaluate and graph a logarithm with a base other than 10 or e, the **Change-of-Base Formula** can be used.

Change-of-Base Formula

$$\log_b x = \frac{\log_a x}{\log_a b}$$

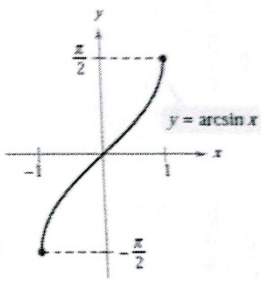
If $a = 10$, then $\log_b x = \frac{\log x}{\log b}$; If $a = e$, then $\log_b x = \frac{\ln x}{\ln b}$.

7.5 The inverse trigonometric functions

- The sine function and the inverse sine function

➢ The Sine Function and the Inverse Sine Function

The domain of $y = \sin x$ is restricted to $[-\frac{\pi}{2}, \frac{\pi}{2}]$. And the corresponding range is $[-1, 1]$.



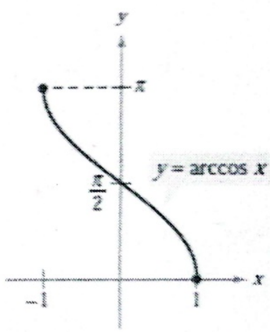
The inverse of $y = \sin x$ is called the inverse sine function, written $y = \sin^{-1} x$ or $y = \arcsin x$, and is defined as follows: $y = sin^{-1}x$ if and only if $sin\, y = x$ and $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

The domain of $y = \sin^{-1} x$ is $[-1, 1]$, and the range is $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

- The cosine function and the inverse cosine function

➢ The Cosine Function and the Inverse Cosine Function

The domain of $y = \cos x$ is restricted to $[0, \pi]$. And the corresponding range is $[-1, 1]$



The inverse of $y = \cos x$ is called the inverse cosine function, written $y = \cos^{-1} x$ or $y = \arccos x$, and is defined as follows: $y = cos^{-1}x$ if and only if $cos\, y = x$ and $0 \le y \le \pi$

The domain of $y = \cos^{-1} x$ is $[-1, 1]$, and the range is $[0, \pi]$.

- The tangent function and the inverse tangent function

8.2 Sine and cosine curves

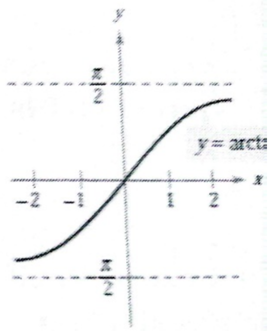
- Frequency: The frequency of a sinusoidal function is the number of cycles that its graph completes in an interval of 2π radians.

- Amplitude: The amplitude of a sinusoidal function is half the difference between its maximum and minimum values.

- Midline: The midline of the graph of a sinusoidal function is determined by the average of the maximum and minimum values of the function.

➢ The Tangent Function and Inverse Tangent Function

The domain of $y = \tan x$ is restricted to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, and the corresponding range is $(-\infty, +\infty)$.



The inverse of $y = \tan x$ is written $y = \tan^{-1} x$ or $y = \arctan x$, and is defined as: $y = tan^{-1}x$ if and only if $tan\, y = x$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$

The domain of $y = \tan^{-1}x$ is $(-\infty, +\infty)$, and the range is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

- Horizontal dilation: y = sin(bx) and y = cos(bx)

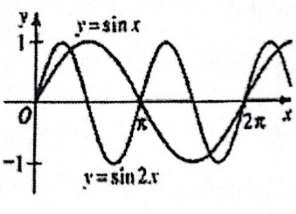
Horizontal Shrinking: $|c| > 1$

eg: $y = \sin 2x$

Amplitude: 1

Period: π

Frequency: $\frac{1}{\pi}$



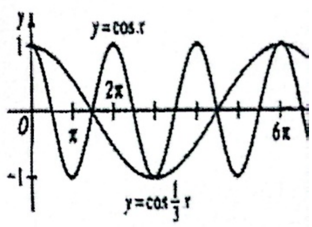
Horizontal Stretching: $|c| < 1$

eg: $y = \cos\frac{1}{3}x$

Amplitude: 1

Period: 6π

Frequency: $\frac{1}{6\pi}$



- Vertical dilation: y = a sinx and y = a cosx

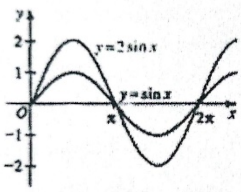
Vertical Stretching: $|a| > 1$

eg: $y = 2\sin x$

Amplitude: 2

Period: 2π

Frequency: $\frac{1}{2\pi}$



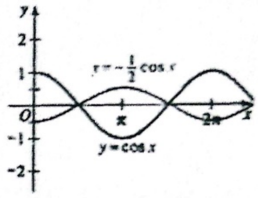
Vertical Shrinking: $|a| < 1$

eg: $y = -\frac{1}{2}\cos x$

Amplitude: $\frac{1}{2}$

Period: 2π

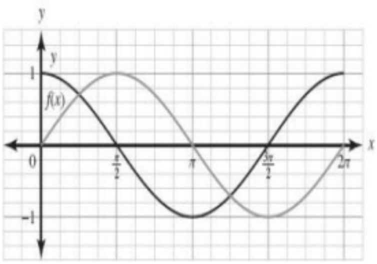
Frequency: $\frac{1}{2\pi}$



- Horizontal translation of y = sin(x + c) and y = cos(x + c)

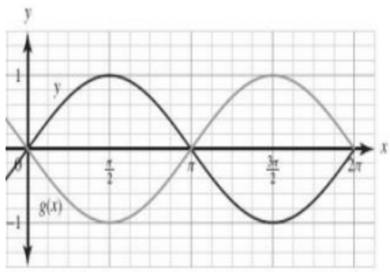
Translate left: $c > 0$

eg: $y = \cos(x - \frac{\pi}{2})$



Translate right: $c < 0$

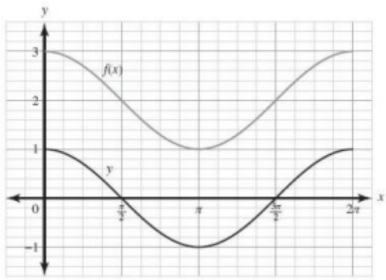
eg: $y = \sin(x + \pi)$



- Vertical Translation of y = sinx + d and y = cosx + d

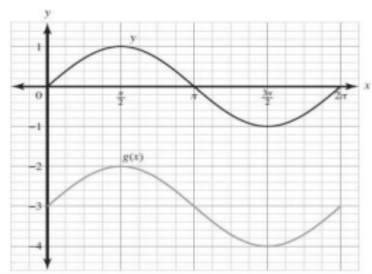
Translate upwards: $d > 0$

eg: $y = \cos x + 2$



Translate downwards: $d < 0$

eg: $y = \sin x - 3$

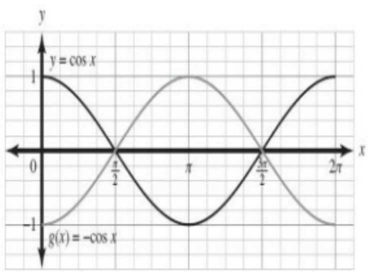


- Reflecting trigonometric function

➢ Reflecting Trigonometric Function

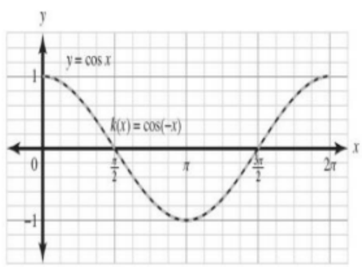
Reflecting on x-axis: $-f(x)$

eg: $y = -\cos x$



Reflecting on y-axis: $f(-x)$

eg: $y = \cos(-x)$



8.3 Modeling periodic behavior

- The smallest interval of input values over which the maximum or minimum output values start to repeat can be used to determine or estimate the period and frequency for a sinusoidal function model.

- The maximum and minimum output values can be used to determine the amplitude and vertical shift for a sinusoidal function model.

- Flexibly select sine or cosine functions based on different situation, making the equation become easier to be calculated as well as promoting critical thinking.

8.4 Relationships among the functions

FUNDAMENTAL TRIGONOMETRIC IDENTITIES

Reciprocal Identities

$$\sin u = \frac{1}{\csc u} \qquad \cos u = \frac{1}{\sec u} \qquad \tan u = \frac{1}{\cot u}$$

$$\csc u = \frac{1}{\sin u} \qquad \sec u = \frac{1}{\cos u} \qquad \cot u = \frac{1}{\tan u}$$

Quotient Identities

$$\tan u = \frac{\sin u}{\cos u} \qquad \cot u = \frac{\cos u}{\sin u}$$

Pythagorean Identities

$$\sin^2 u + \cos^2 u = 1 \qquad 1 + \tan^2 u = \sec^2 u \qquad 1 + \cot^2 u = \csc^2 u$$

Cofunction Identities

$$\sin\left(\frac{\pi}{2} - u\right) = \cos u \qquad \cos\left(\frac{\pi}{2} - u\right) = \sin u$$

$$\tan\left(\frac{\pi}{2} - u\right) = \cot u \qquad \cot\left(\frac{\pi}{2} - u\right) = \tan u$$

$$\sec\left(\frac{\pi}{2} - u\right) = \csc u \qquad \csc\left(\frac{\pi}{2} - u\right) = \sec u$$

Even/Odd Identities

$$\sin(-u) = -\sin u \qquad \cos(-u) = \cos u \qquad \tan(-u) = -\tan u$$

$$\csc(-u) = -\csc u \qquad \sec(-u) = \sec u \qquad \cot(-u) = -\cot u$$

Example 3: Simplify each expression:

(1) $\frac{\tan x+\cot x}{\sec^2 x}$

$= (\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}) \cdot \cos^2 x$

$= \frac{\sin^2 x + \cos^2 x}{\cos x \cdot \sin x} \cdot \cos^2 x$

$= \frac{\cos^2 x}{\cos x \cdot \sin x}$

$= \cot x$

(2) $(\sec^2 x - 1)(\csc^2 x - 1)$

$= \sec^2 x \cdot \csc^2 x - (\sec^2 x + \csc^2 x) + 1$

$= \frac{1}{\cos^2 x} \cdot \frac{1}{\sin^2 x} - (\frac{1}{\cos^2 x} + \frac{1}{\sin^2 x}) + 1$

$= \frac{1}{\cos^2 x \cdot \sin^2 x} - \frac{\sin^2 x + \cos^2 x}{\cos^2 x + \sin^2 x} + 1$

$= 1$

(3) $\frac{\tan^2 x}{\sec x+1} + 1$

$= \frac{\tan^2 x + \sec x + 1}{\sec x + 1}$

$= \frac{\sec^2 x + \sec x}{\sec x + 1}$

$= \sec x$

(4) $\frac{\sec x+\csc x}{1+\tan x}$

$= (\frac{1}{\sin x} + \frac{1}{\cos x}) \cdot \frac{1}{1+\tan x}$

$= \frac{\sin x + \cos x}{\sin x \cdot \cos x} \cdot \frac{\cos x}{\cos x + \sin x}$

$= \frac{1}{\sin x}$

$= \csc x$

(5) $\frac{\sin x \cot x+\cos x}{2\tan(\frac{\pi}{2}-x)}$

$= \frac{\cos x + \cos x}{2\cot x}$

$= \frac{\cos x}{\cot x}$

$= \cos x \cdot \frac{\sin x}{\cos x}$

$= \sin x$

(6) $\frac{\sin^2 x}{1+\cos x}$

$= \frac{1-\cos^2 x}{1+\cos x}$

$= \frac{(1+\cos x)(1-\cos x)}{1+\cos x}$

$= 1-\cos x$

Example 4: Prove $\frac{1}{1-\sec x} + \frac{1}{1+\sec x} = -2\cot^2 x$

Left side. $\frac{1-\sec x+1+\sec x}{(1-\sec x)(1+\sec x)} = \frac{2}{1-\sec^2 x}$

$= -\frac{2}{\sec^2 x - 1}$

$= -\frac{2}{\tan^2 x}$

$= -2\cot^2 x$

$\therefore \frac{1}{1-\sec x} + \frac{1}{1+\sec x} = -2\cot^2 x$

Example 5: Prove the following equation:

(1) $\frac{\cot\theta-\tan\theta}{\sin\theta\cos\theta} = \csc^2\theta - \sec^2\theta$

$\frac{\cot\theta - \tan\theta}{\sin\theta \cdot \cos\theta} = (\frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta}) \cdot \frac{1}{\sin\theta \cdot \cos\theta}$

$= \frac{\cos^2\theta - \sin^2\theta}{\sin^2\theta \cdot \cos^2\theta}$

$= \frac{1}{\sin^2\theta} - \frac{1}{\cos^2\theta}$

$= \csc^2\theta - \sec^2\theta$

(2) $\frac{1-\sin^2\theta}{1+\cot^2\theta} = \sin^2\theta\cos^2\theta$

$\frac{1-\sin^2\theta}{1+\cot^2\theta} = \frac{\sin^2\theta \cdot \cos^2\theta}{\sin^2\theta + \cos^2\theta}$

$= \sin^2\theta \cos^2\theta$

(3) $\frac{\tan^2 x}{1+\tan^2 x} = \sin^2 x$

$\frac{\tan^2 x}{1+\tan^2 x} = \frac{\tan^2 x}{\sec^2 x}$

$= \frac{\sin^2 x}{\cos^2 x} \cdot \cos^2 x$

$= \sin^2 x$

(4) $\frac{\tan x}{1+\sec x} + \frac{1+\sec x}{\tan x} = 2\csc x$

$\frac{\tan x}{1+\sec x} + \frac{1+\sec x}{\tan x} = \frac{\tan^2 x + (1+\sec x)^2}{\tan x(1+\sec x)}$

$= \frac{\frac{\sin^2 x}{\cos^2 x} + \frac{1}{\cos^2 x} + \frac{2}{\cos x} + 1}{\frac{\sin x}{\cos x} + \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x}}$

$= \frac{\sin^2 x + 1 + 2\cos x + \cos^2 x}{\sin x \cdot \cos x + \sin x}$

$= \frac{2\cos x + 2}{\text{[illegible]}} = \frac{2(\cos x + 1)}{\text{[illegible]}} = 2\csc x$

Practice:

(1) $2\sec^2\theta = \frac{1}{1-sin\theta} + \frac{1}{1+sin\theta}$

(2) $tan\theta - 1 = \frac{sec\theta - csc\theta}{csc\theta}$

(3) $\frac{cos\theta}{1+sin\theta} = sec\theta - tan\theta$

(4) $\frac{1}{csc\theta+cot\theta} = \frac{1-cos\theta}{sin\theta}$

(5) $\frac{1-cos\theta}{1+cos\theta} = (csc\theta - cot\theta)^2$

(6) $\frac{1+sin\theta}{cos\theta} + \frac{cos\theta}{1+sin\theta} = \frac{2}{cos\theta}$

(1) $\frac{1}{1-\sin\theta} + \frac{1}{1+\sin\theta} = \frac{2}{1-\sin^2\theta}$

$= \frac{2}{\cos^2\theta}$

$= 2\sec^2\theta$

(2). $\frac{\sec\theta - \csc\theta}{\csc\theta} = \frac{\sec\theta}{\csc\theta} - 1$

$= \frac{\sin\theta}{\cos\theta} - 1$

$= \tan\theta - 1$

(3). $\frac{\cos\theta}{1+\sin\theta} = \frac{\cos\theta(1-\sin\theta)}{\cos^2\theta}$

$= \frac{1-\sin\theta}{\cos\theta}$

$\sec\theta - \tan\theta = \frac{1}{\cos\theta} - \frac{\sin\theta}{\cos\theta}$

$= \frac{1-\sin\theta}{\cos\theta}$

$\therefore \frac{\cos\theta}{1+\sin\theta} = \sec\theta - \tan\theta$

(4) $\frac{1}{\csc\theta + \cot\theta} = \frac{1}{\frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta}}$

$= \frac{\sin\theta}{1+\cos\theta}$

$= \frac{\sin\theta(1-\cos\theta)}{\sin^2\theta}$

$= \frac{1-\cos\theta}{\sin\theta}$

(5) $(\csc\theta - \cot\theta)^2 = \csc^2\theta - 2\csc\theta\cot\theta + \cot^2\theta$

$= \frac{1}{\sin^2\theta} - 2\cdot\frac{1}{\sin\theta}\cdot\frac{\cos\theta}{\sin\theta} + \frac{\cos^2\theta}{\sin^2\theta}$

$= \frac{1+\cos^2\theta}{\sin^2\theta} - \frac{2\cos\theta}{\sin^2\theta}$

$= \frac{(\cos\theta-1)^2}{\sin^2\theta}$

$= \frac{(\cos\theta-1)^2}{(1-\cos\theta)(1+\cos\theta)}$

$= \frac{1-\cos\theta}{1+\cos\theta}$

(6) $\frac{1+\sin\theta}{\cos\theta} + \frac{\cos\theta}{1+\sin\theta} = \frac{(1+\sin\theta)^2 + \cos^2\theta}{\cos\theta(1+\sin\theta)}$

$= \frac{1+2\sin\theta+\sin^2\theta+\cos^2\theta}{\cos\theta+\sin\theta\cos\theta}$

$= \frac{2+2\sin\theta}{\cos\theta(1+\sin\theta)}$

$= \frac{2}{\cos\theta}$

8.5 Solving more difficult trigonometric equations

➢ Solve Trigonometric Equation with Identities

Example 2: Solve for $0 \le \theta < 2\pi$. Give the answers to the nearest hundredth of a radian when necessary.

(1) $2\sec^2 x + \tan x = 5$

$2(1+\tan^2 x)+\tan x = 5$

$2\tan^2 x + \tan x - 3 = 0$

$(2\tan x + 3)(\tan x - 1) = 0$

1) $\tan x = -\frac{3}{2} \to x_1 = 0.69\pi,\ x_2 = 1.69\pi$

2) $\tan x = 1 \to x_3 = \frac{\pi}{4},\ x_4 = \frac{5\pi}{4}$

(2) $\cos^2 x - 3\sin x = 3$

$1 - \sin^2 x - 3\sin x - 3 = 0$

$\sin^2 x + 3\sin x + 2 = 0$

$(\sin x + 1)(\sin x + 2) = 0$

1) $\sin x = -1 \to x_1 = \frac{3\pi}{2}$

2) $\sin x = -2 \to$ reject

$\therefore x = \frac{3\pi}{2}$

(3) $2\cot^2 x + 3\csc x = 1$

$\frac{2\cos^2 x}{\sin^2 x} + \frac{3}{\sin x} = 1$

$2\cos^2 x + 3\sin x - \sin^2 x = 0$

$2 - 2\sin^2 x + 3\sin x - \sin^2 x = 0$

$-3\sin^2 x + 3\sin x + 2 = 0$

$3\sin^2 x - 3\sin x - 2 = 0$

$\sin x = \frac{3 \pm \sqrt{33}}{6}$

1) $\sin x = \frac{3+\sqrt{33}}{6}$, reject

2) $\sin x = \frac{3-\sqrt{33}}{6}$, $x_1 = 1.85\pi,\ x_2 = 1.15\pi$

(4) $\csc^2 x - 2\csc x = 2 - 4\sin x$

$\frac{1}{\sin^2 x} - \frac{2}{\sin x} = 2 - 4\sin x$

$1 - 2\sin x = 2\sin^2 x - 4\sin^3 x$

$4\sin^3 x - 2\sin^2 x - 2\sin x + 1 = 0$

$2\sin^2 x(2\sin x - 1) - (2\sin x - 1) = 0$

$(2\sin^2 x - 1)(2\sin x - 1) = 0$

1) $2\sin^2 x - 1 = 0$

$\sin^2 x = \frac{1}{2}$

a) $\sin x = \frac{\sqrt{2}}{2} \to x_1 = \frac{\pi}{4},\ x_2 = \frac{3\pi}{4}$

b) $\sin x = -\frac{\sqrt{2}}{2} \to x_3 = -\frac{\pi}{4},\ x_4 = \frac{5\pi}{4}$

2) $2\sin x - 1 = 0$

$\sin x = \frac{1}{2}$

$\to x_5 = \frac{\pi}{6},\ x_6 = \frac{5\pi}{6}$

(5) $1 - \sin x = \cos x$

$(1 - \sin x)^2 = \cos^2 x$

$\sin^2 x - 2\sin x + 1 = \cos^2 x$

$\sin^2 x - \cos^2 x + \sin^2 x + \cos^2 x - 2\sin x = 0$

$2\sin^2 x - 2\sin x = 0$

$\sin x(\sin x - 1) = 0$

1) $\sin x = 0 \to x_1 = 0,\ x_2 = \pi \to$ reject

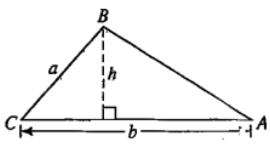
2) $\sin x = 1 \to x_3 = \frac{\pi}{2}$

9.2 The area of a triangle

➢ The Area of a Triangle

The area K of ΔABC is given by:

$$K = \frac{1}{2}\cdot(\text{one side})\cdot(\text{another side})\cdot(\text{sine of included angle})$$



9.3 The law of sines

➢ **The Law of Sines** In ΔABC, $\dfrac{\sin A}{a}=\dfrac{\sin B}{b}=\dfrac{\sin C}{c}$

-- SSA situation is called the **ambiguous case**.

When you are given the lengths of two sides of a triangle and the measure of a nonincluded angle (SSA), it may be possible to construct no triangle, one triangle, or two triangles.

Example 1 Solve ΔRST if $\angle S=40°$, $r=30$, and $s=20$. Give angle measures to the nearest tenth of a degree and lengths to three significant digits.

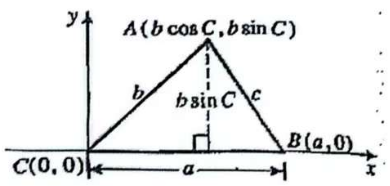
9.4 The law of cosines

➢ **The Law of Cosines**

-- In ΔABC,

$$c^2=a^2+b^2-2ab\cos C$$
$$a^2=b^2+c^2-2bc\cos A$$
$$b^2=a^2+c^2-2ac\cos B$$

$$\begin{pmatrix}\text{side}\\ \text{opposite}\\ \text{angle}\end{pmatrix}^2=\begin{pmatrix}\text{side}\\ \text{adjacent}\\ \text{to angle}\end{pmatrix}^2+\begin{pmatrix}\text{other side}\\ \text{adjacent}\\ \text{to angle}\end{pmatrix}^2-2\begin{pmatrix}\text{one}\\ \text{adjacent}\\ \text{side}\end{pmatrix}\begin{pmatrix}\text{other}\\ \text{adjacent}\\ \text{side}\end{pmatrix}\cos(\text{angle})$$



$$\begin{aligned}c^2=(AB)^2&=(b\cos C-a)^2+(b\sin C-0)^2\\&=b^2\cos^2 C-2ab\cos C+a^2+b^2\sin^2 C\\&=b^2(\cos^2 C+\sin^2 C)-2ab\cos C+a^2\\&=b^2\cdot 1-2ab\cos C+a^2\\&=a^2+b^2-2ab\cos C\end{aligned}$$

-- To solve the angle $\cos C=\dfrac{a^2+b^2-c^2}{2ab}$ $\qquad \cos A=\dfrac{b^2+c^2-a^2}{2bc}$ $\qquad \cos B=\dfrac{a^2+c^2-b^2}{2ac}$

$$\cos(\text{angle})=\frac{(\text{adjacent})^2+(\text{adjacent})^2-(\text{opposite})^2}{2\cdot(\text{adjacent})\cdot(\text{adjacent})}$$

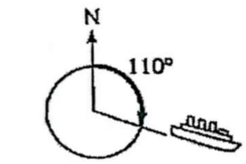
Summary:

Given:	Use:	To find:
SAS	Law of cosine	The third side and then one of the remaining angles.
SSS	Law of cosine	Any two angles.
ASA or AAS	Law of sine	The remaining sides.
SSA	Law of sine	An angle opposite a given side and then the third side. (Note that 0, 1 or 2 triangles are possible.)

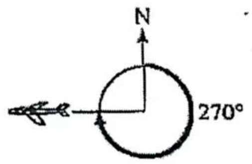
9.5 Applications of trigonometry to navigation and surveying

- **Course**

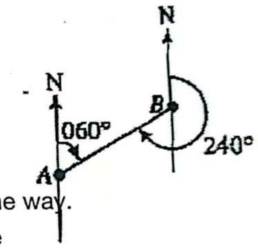
-- The **course** of a ship or plane is the angle, measured In degree, from the clockwise direction to the direction of the ship or plane.



course of ship = 110°



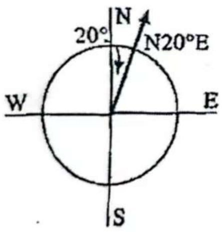
course of plane = 270°

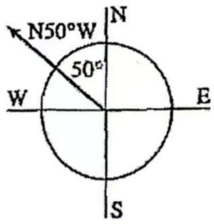


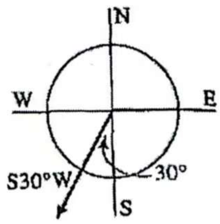
-- The **compass bearing** of one location from another is measured in the same way.

-- In surveying, a compass reading is usually given as an acute angle from the north-south line toward the east or west.

bearing of B from A = 060°
bearing of A from B = 240°



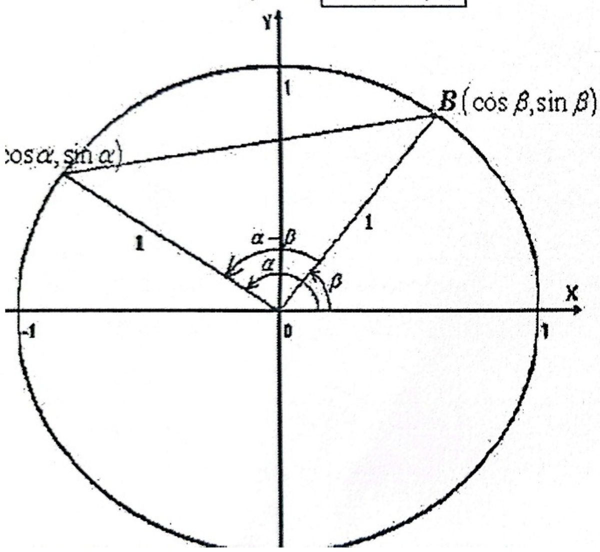




10.1 Formulas for cos(A + B) and sin(A + B)

Formulas for $\cos(\alpha \pm \beta)$

Prove: The identity for $\boxed{\cos(\alpha - \beta)}$:



$AB^2 = 2 - \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$\therefore \cos(\alpha - \beta) = \dfrac{OA^2 + OB^2 - AB^2}{2OA \cdot OB}$

$= \dfrac{2 - (2 - \cos\alpha\cos\beta - \sin\alpha\sin\beta)}{2}$

$= \cos\alpha\cos\beta + \sin\alpha\sin\beta$

Q1: How about the case that $\alpha - \beta < 0$? Does th
identity hold? Why?

Q2: How about $\cos(\alpha + \beta)$?

$\cos(\alpha + \beta) = \cos\alpha\cos(-\beta) + \sin\alpha\sin(-\beta)$

$= \cos\alpha\cos\beta - \sin\alpha\sin\beta$

Q3: Could you find the formulas for $\sin(a \pm \beta)$?

$\sin(\alpha+\beta) = \cos(\frac{\pi}{2} - \alpha - \beta)$

$= \cos(\frac{\pi}{2} - \alpha)\cos\beta + \sin(\frac{\pi}{2} - \alpha)\sin\beta$

$= \sin\alpha\cos\beta + \cos\alpha\sin\beta$

$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

-- Sum and Difference Formulas for Cosine and Sine

$\cos(a+\beta) = \cos a\cos\beta$ __$-$__ $\sin a\sin\beta$	$\sin(a+\beta) = \sin a\cos\beta$ __$+$__ $\cos a\sin\beta$
$\cos(a-\beta) = \cos a\cos\beta$ __$+$__ $\sin a\sin\beta$	$\sin(a-\beta) = \sin a\cos\beta$ __$-$__ $\cos a\sin\beta$

Form: $a\sin x \pm b\cos x$

1) $a\sin x + b\cos x = \sqrt{a^2+b^2}\sin(x+\phi)$, where $\sin\phi = \frac{b}{\sqrt{a^2+b^2}}$ and $\cos\phi = \frac{a}{\sqrt{a^2+b^2}}$;

2) $a\sin x + b\cos x = \sqrt{a^2+b^2}\cos(x-\eta)$, where $\sin\eta = \frac{a}{\sqrt{a^2+b^2}}$ and $\cos\eta = \frac{b}{\sqrt{a^2+b^2}}$.

8. Rewrite the function $y = \sqrt{3}\cos x + 3\sin x$ with a single sine or cosine function and identify the amplitud

$y = 2\sqrt{3}(\frac{1}{2}\cos x + \frac{\sqrt{3}}{2}\sin x)$

$= 2\sqrt{3}(\sin\frac{\pi}{6}\cos x + \cos\frac{\pi}{6}\sin x)$

$= 2\sqrt{3}\sin(x+\frac{\pi}{6})$

$y = 2\sqrt{3}(\frac{1}{2}\cos x + \frac{\sqrt{3}}{2}\sin x)$

$= 2\sqrt{3}(\cos\frac{\pi}{3}\cos x + \sin\frac{\pi}{3}\sin x)$

$= 2\sqrt{3}\cos(x-\frac{\pi}{3})$

10.2 Formulas for tan(A + B)

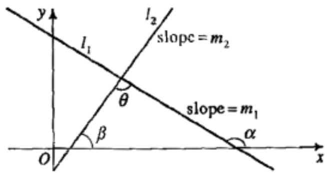
➢ **Sum Formula for Tangent** $\tan(a+\beta) = \frac{\tan\alpha + \tan\beta}{1-\tan\alpha\tan\beta}$

Prove: $\tan(a+\beta) = \frac{\sin(a+\beta)}{\cos(a+\beta)} = \frac{\_\_\_\_\_\_\_\_}{\_\_\_\_\_\_\_\_} \div \cos a\cos\beta = \frac{\_\_\_\_\_\_\_\_}{\_\_\_\_\_\_\_\_}$

$= \_\_\_\_\_\_\_\_\_\_\_\_$

➢ **Difference Formula for Tangent** $\tan(a-\beta) = \frac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta}$

- **Angle between Two Lines** The formula for $\tan(\alpha-\beta)$ can be used to find an angle between two intersecting lines. Consider the angle θ between lines l_1 and l_2 with slopes m_1 and m_2. The inclination α of $l_1: \tan\alpha = m_1$ and the inclination β of $l_2: \tan\beta = m_2$. According to the graph, $\alpha = \beta + \theta$ and $\theta = \alpha - \beta$.



Therefore, $\boxed{\tan\theta = \tan(\alpha-\beta) = \dfrac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta} = \dfrac{m_1 - m_2}{1+m_1 m_2}}$

10.3 Double-angle and Half-angle formula

- **Double-Angle Formulas**

$\sin 2\alpha = 2\sin\alpha\cos\alpha$ $\qquad$ $\tan 2\alpha = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$

$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 1-2\sin^2\alpha = 2\cos^2\alpha - 1$

- **Half-Angle Formulas**

$\sin\dfrac{\alpha}{2} = \pm\sqrt{\dfrac{1-\cos\alpha}{2}}$ $\qquad$ $\cos\dfrac{\alpha}{2} = \pm\sqrt{\dfrac{1+\cos\alpha}{2}}$

$\tan\dfrac{\alpha}{2} = \pm\sqrt{\dfrac{1-\cos\alpha}{1+\cos\alpha}} = \dfrac{\sin\alpha}{1+\cos\alpha} = \dfrac{1-\cos\alpha}{\sin\alpha}$

10.4 Solving trigonometric equations

- Equivalent trigonometric forms may be useful in solving trigonometric equations. 10.4.A.3 Because trigonometric functions are periodic, there are often infinitely many solutions to trigonometric equations. 10.4.A.4 Algebraic and graphical methods are used to solve trigonometric equations in a finite interval, including the use of trigonometric identities and factorization.

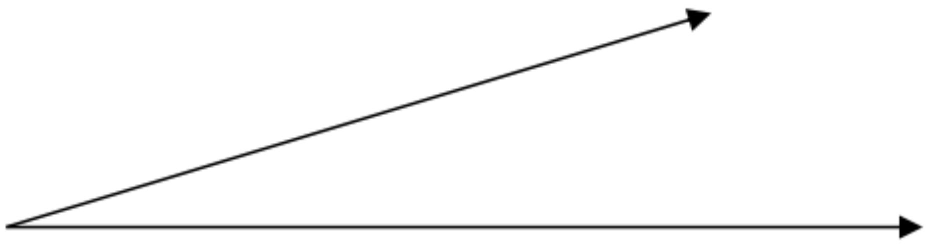
- Inverse trigonometric functions are useful in solving equations involving trigonometric functions, but solutions may need to be modified due to domain restrictions. 10.4.A.6 In trigonometric equations arising from a contextual scenario, there is often a domain restriction that can be implied from the context, which limits the number of solutions.

- Equivalent trigonometric forms may be useful in solving trigonometric inequalities. 10.4.B.2 Inverse trigonometric functions are useful in solving inequalities involving trigonometric functions, but solutions may need to be modified due to domain restrictions. 10.4.B.3 In trigonometric inequalities arising from a contextual scenario, there is often a domain restriction that can be implied from the context, which limits the number of solutions.

11.1 Polar coordinates and graphs

- Polar coordinates

The polar coordinate system is an alternative to the Cartesian system of rectangular coordinates for locating points in a plane. λ The polar coordinates of a point P in the polar coordinate system consist of an ordered pair (r,θ). The directed distance from the pole to P is r, and the measure of the angle from the polar axis to OP is θ . The polar coordinates of the pole O are (0,θ) . Angle θ can be measured in degrees or radians. Both r and θ can be either positive or negative. When θ is positive, the polar angle is obtained by rotating OP counterclockwise from the polar axis, and when θis negative, the polar angle is obtained by rotating OP clockwise from the polar axis. When r is positive, the polar distance is measured from O along the terminal side of the angle θ , and When r is negative, the polar distance is measured from O on the ray opposite the terminal side of the angle θ. A plane is called an rθ -plane when polar coordinates (r,θ) are used to identify its points.



13.1 Arithmetic and geometric sequences

Means

<Arithmetic and Geometric Means>

- Arithmetic Mean:

  If the numbers $a_1, a_2, a_3, \ldots, a_{k-1}, a_k$ form an arithmetic sequence, then $a_2, a_3, \ldots, a_{k-1}$ are ***arithmetic means*** between a_1 *and* a_k.

- Geometric Mean:

  A single geometric mean between two numbers is called the ***geometric mean***, or ***mean proportional***, between the two numbers. It is customary to let mean proportional $m = \sqrt{ab}$ when a and b are both positive, $m = -\sqrt{ab}$ when a and b are both negative. When a and b have opposite signs, there is no real mean proportional.

Definitions

<Definitions of a sequence>

- Sequence: A function whose domain is the set of positive integers (**infinite sequence**) or a subset of the positive integers that consists of the n integers, for example: 1,2,3,…..,n (**finite sequence**).

- Terms: The numbers in the range of the sequence function

In brief, a sequence is a set of numbers, called terms, arranged in a particular **order**.

<Notations>

Sequence $\{a_n\}$ Terms: $a_1, a_2, a_3, \ldots\ldots$ General term: a_n

Sequence $\{f(n)\}$ Terms: $f(1), f(2), f(3), \ldots\ldots$ General term: $f(n)$

<Rules of Sequences>

- Rules given **explicit**ly:

  all terms can be found **directly**, using the explicit formula

- Rules given **recursive**ly:

  given **enough** initial term(s), define a_n using preceding terms in a recursive formula

<Two simplest types of sequences>

- Arithmetic Sequence

A sequence $a_1, a_2, a_3, \ldots$ is an ***arithmetic sequence*** or arithmetic progression if there is a constant d for which $a_n = a_{n-1} + d$ for all integers $n > 1$. The constant d represents the ***common difference*** of the sequence, and $d = a_n - a_{n-1}$ for all integers $n > 1$. $a_n = a_1 + (n-1)d$

- Geometric Sequence

A sequence $a_1, a_2, a_3, \ldots$ is a ***geometric sequence***, or a geometric progression, if there is a constant r for which $a_n = a_{n-1} \cdot r$ for all integers n>1. The constant r is called the ***common ratio*** of the sequence, and $r = \dfrac{a_n}{a_{n-1}}$, for all integers $n > 1$. $a_n = a_1 \cdot r^{n-1}$

13.3 Arithmetic and geometric series and their sums

<Definitions>

- Sequence Vs. Series

Finite sequence: 2, 6, 10, 14

Finite series: 2 + 6 + 10 + 14

Infinite sequence: $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots, \frac{1}{2^n}, \cdots$

Infinite series: $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots+\frac{1}{2^n}+\cdots$

<Notations>

S_n represents the sum of n terms of a series, and it is also called "**partial sum**"

- S_n given **explicit**ly for the sequence $\{t_n\}$:

$$S_n = t_1 + t_2 + t_3 + \cdots + t_n;$$

- S_n given **recursive**ly for the sequence $\{t_n\}$:

$$\begin{cases} S_1 = t_1 \end{cases} \qquad \begin{cases} S_0 = 0 \end{cases}$$

o

<Sum of a Finite **Arithmetic Series**>

$$S_n = \frac{(a_1 + a_n)n}{2} = na_1 + \frac{n(n-1)}{2}d$$

e

<Sum of a Finite **Geometric Series**>

$$S_n = \begin{cases} \frac{a_1(1-r^n)}{1-r} & (\qquad) \\ & \end{cases}$$

13.4 Limits of infinite sequences

<Definitions>

- A sequence that does not have a last term is called **infinite.**

For example:

the sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \cdots, \frac{1}{n}, \cdots$ is infinite, and it is known as the **harmonic sequence**;

the sequence $\left\{\frac{1}{e^n}\right\}$ is infinite, if n is allowed to take all the values in $\mathbb{Z}^+$.

- Limit of an infinite sequence $\{u_n\}$:

$$\lim_{n\to\infty} u_n$$

- If the limit of the infinite sequence exists, which means the limit is a real number, then the sequence is **convergent**, or otherwise is **divergent**.

<Some basic rules>

If L, M, and k are **real numbers**, and $\lim_{n\to\infty} a_n = L$, $\lim_{n\to\infty} b_n = M$

- $\lim_{n\to\infty}\left(a_n \pm b_n\right) = L \pm M$
- $\lim_{n\to\infty}\left(ka_n\right) = kM$
- $\lim_{n\to\infty}\left(a_n \cdot b_n\right) = L \cdot M$
- If $M \neq 0$, then $\lim_{n\to\infty}\left(\dfrac{a_n}{b_n}\right) = \dfrac{L}{M}$
- If function $f(x)$ is continuous and defined everywhere, then $\lim f(a_n) = f(L)$.

<The Squeeze Theorem> (also known as the Sandwich Theorem)

If for all $n > N_0$, where N_0 is some positive integer,

$$a_n \leq c_n \leq b_n,$$

and

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = L \in \mathbb{R},$$

then

$$\lim_{n\to\infty} c_n = L.$$

13.5 Sums of infinite series

<Sum of an Infinite Geometric Series>

If $|r| < 1$, the infinite geometric series

$$a_1 + a_1 r + a_1 r^2 + \cdots + a_1 r^n + \cdots$$

converges to the sum $S = \dfrac{a_1}{1-r}$.

If $|r| \geq 1$, and $a_1 \neq 0$, then the series diverges.

<Power Series>

An expression of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots$$

is a **power series centered at $x = 0$.** An expression of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n + \cdots$$

is a **power series centered at $x = a$.**

- The term $c_n(x-a)^n$ is the ***n*-th term**;
- A set of **all** the values of x such that the power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ converges is called the **interval of convergence**.

13.6 Sigma notation

<Sigma Notation>

The Greek letter $\sum$ (sigma) is often used to express a series or its sum in abbreviated form. For example,

$$\sum_{k=3}^{100} k^2 = 3^2 + 4^2 + 5^2 + \cdots + 100^2 .$$

It may be read as "the sum of k^2 for values of k from 3 to 100."

The k^2 is called the **summand**, the numbers 3 (lower limit) and 100 (upper limit) are called the **limits of summation**, and the symbol k is called the **index**.

<Properties of Finite Sums>

- $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$.
- $\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$

<Infinite Series>

For any infinite series $t_1 + t_2 + t_3 + \cdots + t_n + \cdots = \sum_{n=1}^{\infty} t_n$,

$$S_n = t_1 + t_2 + t_3 + \cdots + t_n$$

is called the ***n*th partial sum**. If the sequence of partial sums S_1, S_2, …, S_n, … has a finite limit S, the infinite series is said to **converge** to the sum S,

$$S = \lim_{n\to\infty} S_n .$$

<Power Series>

An expression of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots$$

is a **power series centered at $x = 0$.** An expression of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n + \cdots$$

is a **power series centered at $x = a$.**

- The term $c_n(x-a)^n$ is the ***n*-th term**;

1 3 . 7 ■ A set of **all** the values of x such that the power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ converges is called the **interval of convergence**.

Mathematica induction

<Mathematical Induction>

- ***The principle of Mathematical Induction***

Let P_n be a statement involving the positive integer n. The statement is true for every $n \geq 1$, if the following two conditions hold.

1. The statement is true for $n = 1$, that is , P_1 is true.
2. For any positive integer k, whenever P_k is true, then P_{k+1} is true.